

Sequence Alignment Theory and Applications

Learning Outcomes

At the end of this module, participants should be able to:

- ✧ demonstrate basic applications of sequence alignment,
- ✧ explain the concepts of homology, identity, orthologues and paralogues,
- ✧ identify point mutations, deletions, insertions in aligned sequences and
- ✧ differentiate between exhaustive and heuristic approach to sequence alignment,

Learning Objectives

- To discuss the concepts of sequence alignment and its applications
- To explain the concepts of homology, identity, orthologues and paralogues
- Demonstrate basic sequence alignment, introducing concepts of point mutations, deletions and insertions
- To examine the different approaches to sequence alignment - exhaustive vs heuristic

Introduction

- ❖ Comparison is one of the most effective methods used in making discovery in Science
- ❖ The most common comparative method used in molecular biology is sequence alignment

What is Sequence Alignment?

- ❖ Is a way of arranging two or more sequences of DNA, RNA, or protein to identify regions of similarity
- ❖ Comparing two sequences – Pairwise alignment
- ❖ Comparing more than 2 sequences – Multiple sequence alignment

Why Align Sequences?

- ❖ To predict function of gene or protein sequence
- ❖ To merge short strings of DNA or RNA sequences
- ❖ To automatically annotate newly sequenced genomes
- ❖ It also the first step in phylogenetic analysis
- ❖ Essential for molecular structure prediction
- ❖ To detect gene mutations – substitution, insertions/deletions

Alignment sequences

- Sequence alignment of nucleotide or amino acid residues are typically represented as rows and columns

Column
↓
Row → 1 AGCTGGCATTATGGATGGCTG
2 AGCTGGCATTATGGATGGCTG

Similarity and Identity - Nucleotides

- ❖ Apparent likeness or resemblance between sequences

1 AGCTGGCATTATGGATGGCTG
2 AGCTGGCATTATGGATGGCTG

Identical sequences
100% Identity
100% Similarity

- ❖ It is an objective repeatable criteria

Comparing Two Sequences

3 AGCTGGCATTATGGATGGCTG
4 AGCTGACATTACGTATGGCTG

90% identity

90% similarity

Point mutations

- ❖ Sequence similarity and sequence identity are synonymous for nucleotide sequences

Percentage of Seq. Similarity/Identity

- ❖ Two sequences of unequal length, percentage of similarity S or identity I

$$I(S) = (L_{i(s)} / L_y) 100$$

Where

$L_{i(s)}$ is the number of aligned residues with similar or identical characteristics

L_y is the length of the shorter of the two sequences

Percentage of Sequence Similarity/Iden

- ❖ Two sequences of equal length, percentage of similarity S or identity I

$$= [2L / (L_y + L_z)] \times 100$$

Where

L is the number of aligned residues with similar or identical characteristics

L_y is the total length of sequence y

L_z is the total length of sequence z

Basis of Sequence Alignment



The overall goal of pairwise sequence alignment is to find the best pairing of two sequences



One sequence needs to be shifted relative to the other to find the position where maximum matches are occur

Comparing two sequences

Which of the following represents optimal alignment of the sequences **AFRICA** and **ARCA**?

- | | | | |
|---|-----------------------|---|---------------------|
| 1 | AFRICA

ARCA | 3 | AFRICA

ARCA |
| 2 | AFRICA

ARCA | 4 | AFRICA

ARCA |

Better Alignment

- ❖ We insert gaps (a base-to-nothing match)

AFRICA
| | |
A-R-CA

Global Alignment

- ❖ Maximises the the number of base-to base matches
- ❖ The order of bases in each of the sequence is maintained
- ❖ Gap-to-gap match are not allowed

Alignment Gaps

Which of the following alignment is more likely optimal alignment for sequences GATTCGGACCTCGGATCCCGT and ATCGACTCGATCGT?

- | | | |
|---|--|--|
| 1 | GATTCGGACCTCGGATCCCGT
-AT-C-GA-CTC-GAT-C-GT | ❖ It is difficult to choose one

❖ = $\sim 2^{39}$ possible alignments for two sequences |
| 2 | GATTCGGACCTCGGATCCCGT
-A-TCG-AC-TC-GAT--CGT | |
| 3 | GATTCGGACCTCGGATCCCGT
-AT-C-GAC-TCG-ATC--GT | |

Search for Optimal Alignment

- ❖ It is desirable to allow some gaps to be introduced into an alignment to compensate for insertions and deletions but this should not be arbitrarily inserted
- ❖ Number of strategies have been proposed for penalising gaps

Optimal Alignment

- ❖ Employ scoring scheme – to reward for matching, punish for mismatching and gaps
- ❖ Similarity of two sequences = high score
- ❖ Dissimilarity of two sequences = low score
- ❖ Account for substitutions, insertions or deletions

Gaps Penalties

- ❖ Constant gap penalty, a penalty of “-a” for any gap, irrespective of the length of the gap.
For high score
 - ✧ The total number of gaps matters
 - ✧ Minimise the number of gap
- ❖ Linear gap penalty, a penalty of “-a” per unit length of a gap
 - ✧ The penalty for one large gap is the same as for many small gaps

Gap Penalties - Continuation

For gap penalty -1	Constant gap penalty	Linear gap penalty
AAAGGCCTTCTAGGGGCTATCC --A-G---TCTA-GG---ATCC	-10	-5
AAAGGCCTTCTAGGGGCTATCC --AG----TCTAGG----ATCC	-10	-3

Gap Penalties - Continuation

- ❖ Constant gap penalty and Linear gap penalty do not consider whether the gap is opening or extending
- ❖ Gaps at the terminal regions are often treated with no penalty because in reality many true homologous sequences are of different lengths

Gap Penalties - Continuation

- ❖ Affine gap penalty is a differential gap penalty for introducing and extending gaps
- ❖ For example, one may use “-y” for gap opening penalty and “-z” for gap extension penalty
 - Total gap penalty, $G = y + (k-1)z$
 - Where k is the length of the gap
- ❖ Drawback, the penalty points (y and z) are arbitrarily chosen constants

Affine Gap Penalty

- ❖ It favours longer gaps over single gaps of the same total length
- ❖ Reflects more likely in biology, one gap of 15 is likely in sequence than 15 small gaps of 1

Alignment Algorithms

- ❖ Different optimisation strategies are used in aligning similar residues
- ❖ Computer algorithm can then determine not just minimal edit distances but optimal alignment
- ❖ Types of algorithms are based on one of these three methods:
 - the exhaustive method,
 - the heuristic method
 - Dot matrix methods

Exhaustive Algorithm

- ❖ Produces list of all possible alignments between two sequences, score and select the alignment with best score (OPTIMAL ALIGNMENT)
- ❖ Dynamic programming (Needleman-Wunsh, 1970)
- ❖ The number of possible global alignments between sequences of length L is $2^{2L}/\sqrt{\pi L}$
- ❖ For sequences of 21 bases/residues, there are $\sim 2^{39}$ possible ways of aligning them

Exhaustive Algorithm

- ❖ Databases continued to increase at an exponential rate, making straight forward application of dynamic programming methods impractical for database searching
- ❖ NOT practical for multiple sequence alignment

Heuristic Algorithm

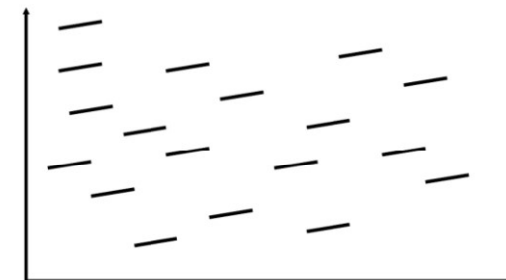
- ❖ Heuristic method is based on the strategy of breaking a sequence up into short runs of consecutive letters called words
- ❖ Makes use of approximations to significantly speed up sequence comparisons
- ❖ Finds a solution close to the best one, fast and easily
- ❖ But with a small risk that true alignment can be missed

Heuristic Algorithms

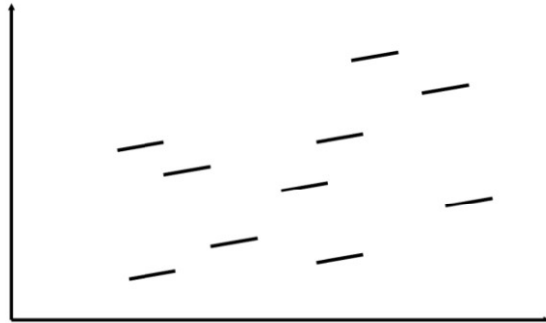
- ❖ Progressive alignment
- ❖ Iterative alignment
- ❖ Block- based alignment
- ❖ Examples of heuristic algorithms - Clustal, Tcoffee, BLAST etc.

Heuristic Algorithms

Block- based alignment is basic form of the heuristic alignment methods



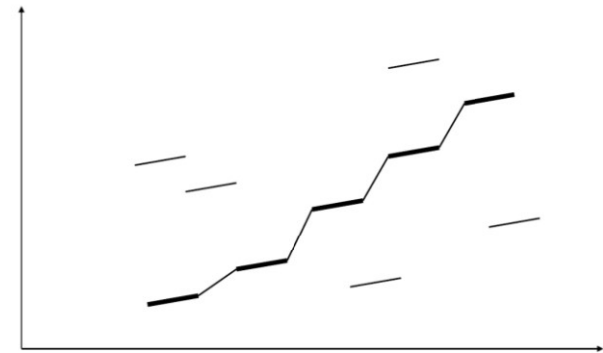
Block Based Method



Iterative Alignment Method

- Realign the initial sequences as well as adding new sequences to the growing multiple sequence alignment

Block Based Method



Progressive Alignment

- ❖ Progressive alignment is a pairwise alignment algorithm is used iteratively, first to align the most closely related pair of sequences, then the next most similar one to that pair and so on
- ❖ The rule “once a gap, always a gap” based on the fact that the positions and lengths of gaps introduced between more similar pairs of sequences should not be affected by more distantly related ones

Pairwise Sequence Alignment

- ❖ Pairwise sequence alignment is the fundamental component of many bioinformatics applications
- ❖ It is extremely useful in structural, functional, and evolutionary analyses of sequences

Global Alignment

- ❖ The two sequences to be aligned are assumed to be generally similar over their entire length
- ❖ Alignment is carried out from beginning to end of both sequences to find the best possible alignment across the entire length between the two sequences
- ❖ More applicable for aligning two closely related sequences of roughly the same length

Disadvantage of Global Alignment

- ❖ May not be able to generate optimal results for divergent sequences and sequences of variable lengths
- ❖ It fails to recognize highly similar local regions between the two sequences

Example (1)

Lets look the alignment of the following sequences **AFRICANISED** and **AFRICNSED**

AFRICANISED
AFRIC-N-SED

Global Alignment

Local Alignment

- ❖ Finds local regions with the highest level of similarity between the two sequences and aligns these regions
- ❖ Used for aligning more divergent sequences with the goal of searching for conserved patterns in DNA or protein sequences
- ❖ The two sequences to be aligned can be of different lengths.
- ❖ The paired subsequences may be surrounded by residues that are completely unrelated

- ❖ More appropriate for aligning divergent biological sequences containing modules, domains or motifs

Example (2)

Lets look the alignment of the following sequences **AFRICANISED** and **FRIC**

AFRICANISED
-FRIC-----

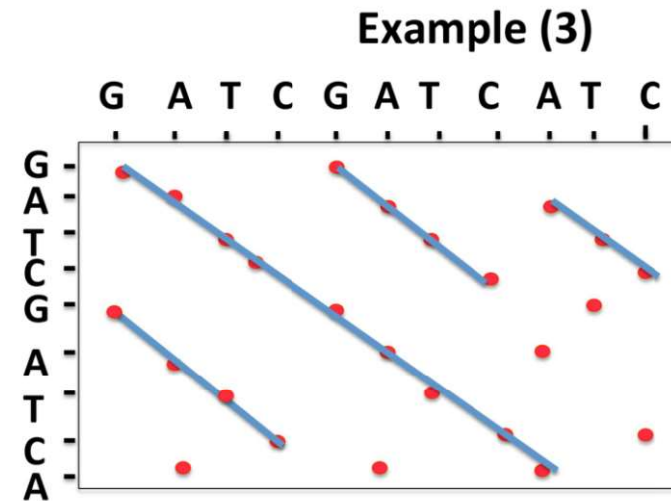
Local Alignment

Alignment Algorithms

- ❖ Both global and local are fundamentally similar
- ❖ Differ in the optimization strategy used in aligning similar residues

Dot-Matrix Representations

- ❖ The most basic sequence alignment method
- ❖ Use the sequences as the coordinates of a two-dimensional graph and then plot points of correspondence within its interior
- ❖ When two sequences are consistently matching over an extended region, the dots will merge to form a diagonal line segment



Dot-Matrix Representations

- ❖ Has ability to reveal complex relationships involving multiple regions of local similarity
- ❖ Gives a direct visual statement of the relationship between two sequences and helps easy identification of the regions of greatest similarities

Advantages of Dot Matrix

- ❖ Easy identification of the regions of greatest similarities
- ❖ Identification of sequence repeat regions based on the presence of parallel diagonals
- ❖ Useful in identifying chromosomal repeats and in comparing gene order conservation between two closely related genomes

Disadvantages of Dot Matrix

- ❖ Lacks statistical rigour in assessing the quality of the alignment
- ❖ Restricted to pairwise alignment
- ❖ It is difficult for the method to scale up to multiple alignment

Optimal Alignment Methods

- ❖ To find the optimal path through a graph
- ❖ Sum the incremental contributions of each step along its route
- ❖ Positive incremental scores used for aligning identical residues, with negative scores used for substitutions and gaps
- ❖ Total score is maximal will give us the best sequence alignment

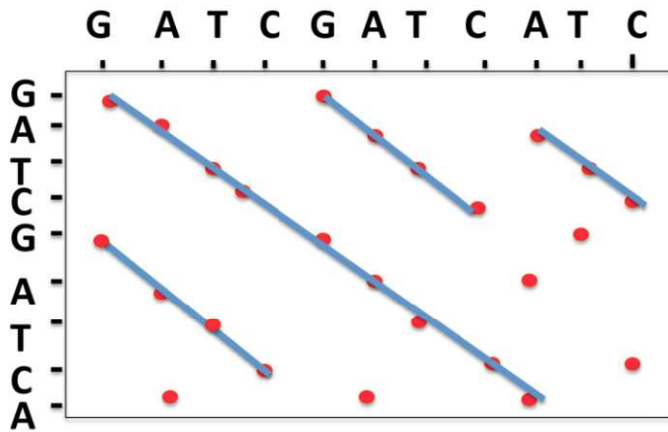
Needleman-Wunsch Algorithm

- ❖ Is an application of a best-path strategy aimed to reduce the massive number of possibilities and still guarantee the best solution will be found (Needleman and Wunsch, 1970)
- ❖ Is an example of dynamic programming
- ❖ Any partial subpath that ends at a point along the true optimal path must itself be the optimal path leading up to that point

Needleman-Wunsch Algorithm (1)

- ❖ The optimal alignment must extend from beginning to end in both sequences
- ❖ It seeks global alignments

Example (3)



How Does Dynamic Programming Work?

- ❖ Break the problem into smaller subproblems
- ❖ Solve the smaller problems optimally
- ❖ Use the subproblem solutions to construct an optimal solution for original problem

Dynamic Algorithm

Consists of three steps

- ❖ Initialisation of score matrix
- ❖ Calculation of scores and filling the traceback matrix
- ❖ Deducing the alignment from the traceback matrix

Example (4)

Find the optimal alignment for the sequences AGATCAT and AGCGT, match +1, mismatch 0 and gap -1

Initialisation Step

Create Matrix with $M + 2$ columns and $N + 2$ rows. The second cell of the second row and the second column filled with 0

		A	G	A	T	C	A	T
	0	-1	-2	-3	-4	-5	-6	-7
A	-1							
G	-2							
C	-3							
G	-4							
T	-5							

Calculation of Scores

Horizontal movement is a gap, -1

Diagonal movement is match, +1 or mismatch, 0

		A	G	A	T	C	A	T
	0	-1	-2	-3	-4	-5	-6	-7
A	-1							
G	-2							
C	-3							
G	-4							
T	-5							

Calculation of Scores

Horizontal movement is a gap, -1

Diagonal movement is match, +1 or mismatch, 0

		A	G	A	T	C	A	T
	0	-1	-2	-3	-4	-5	-6	-7
A	-1	1 or 0	-2					
G	-2							
C	-3							
G	-4							
T	-5							

Calculation of Scores

Horizontal movement is a gap, -1

Diagonal movement is match, +1 or mismatch, 0

		A	G	A	T	C	A	T
	0	-1	-2	-3	-4	-5	-6	-7
A	-1	1	-1	-1	-3	-4	-4	-6
G	-2	-1	0	-1	-2	-3	-2	-5
C	-3	-2	-1	0	-1	0	-1	-4
G	-4	-3	0	1	0	1	0	-3
T	-5	-4	-3	0	1	2	1	-2

Calculation of Scores

Filling the traceback matrix

		A	G	A	T	C	A	T
	0	-1	-2	-3	-4	-5	-6	-7
A	-1	1	0	-1	-2	-3	-4	-5
G	-2	0	2	1	0	-1	-2	-3
C	-3	-1	1	2	1	1	0	-1
G	-4	-2	0	1	2	1	1	0
T	-5	-3	-1	0	2	2	1	2

Traceback Path

- ❖ The optimal alignment score is calculated in the lower-right corner

		A	G	A	T	C	A	T
	0	-1	-2	-3	-4	-5	-6	-7
A	-1	1	0	-1	-2	-3	-4	-5
G	-2	0	2	1	0	-1	-2	-3
C	-3	-1	1	2	1	1	0	-1
G	-4	-2	0	1	2	1	1	0
T	-5	-3	-1	0	2	2	1	2

Traceback

- ❖ Traceback is the process of deducing the best alignment
- ❖ Begins with the last cell filled with the score
- ❖ Then move according to the traceback value written in the cell
- ❖ There are three possible moves – diagonally toward the top-left corner of the matrix, up or left
- ❖ The traceback is completed when the first, top-left cell of the matrix is reached

Optimal Global Alignment

A G A T C A T
 | | | |
 A G - - C G T
 1 +1 -1 -1 +1 +0 +1 = 2

Optimal alignment score = 2

Advantages of Needleman-Wunsch Algorithm

- ❖ Suitable for aligning two closely related sequences that are of the same length
- ❖ But two similar sequences of unequal lengths can be aligned without penalizing terminal gaps

Disadvantage of Needleman-Wunsch Algorithm

- ❖ Risk of missing the best local similarity
- ❖ Divergent sequences or sequences with different domain structures, the approach does not produce optimal alignment

Semi-global Alignment

- ❖ Look for the best alignment between a short and a long sequence
- ❖ Internal and terminal gaps are scored differently as terminal gaps can arise from difference in sequence lengths

Semi-global Alignment (1)

- ❖ Two modifications are made to the basic dynamic programming algorithm
 - ✧ No penalties for the initial gaps – the first row and column of the table are signed with zeros
 - ✧ No penalties at the end gaps – Vertical moves in the last column and horizontal moves in the last row have no gap penalty

Example (5)

Find the optimal alignment for the sequences AGATCAT and GAT, match +1, mismatch 0 and gap -1

Calculation of Scores

❖ Match = +1 , Mismatch = 0 and Gap = -1

		A	G	A	T	C	A	T
	0	0	0	0	0	0	0	0
G	0	0	1	0	0	0	0	0
A	0	1	0	2	1	0	1	0
T	0	0	1	1	3	3	3	3

Initialisation Step

- ❖ Create Matrix with $M + 2$ columns and $N + 2$ rows.
- ❖ The first row and column of the table are signed with zeros

		A	G	A	T	C	A	T
	0	0	0	0	0	0	0	0
G	0							
A	0							
T	0							

Traceback Path

- ❖ The optimal alignment score is the one with the highest score

		A	G	A	T	C	A	T
	0	0	0	0	0	0	0	0
G	0	0	1	0	0	0	0	0
A	0	1	0	2	1	0	1	0
T	0	0	1	1	3	3	3	3

Optimal Semi-global Alignment

A	G	A	T	C	A	T
-	G	A	T	-	-	-

$$0 + 1 + 1 + 1 + 0 + 0 + 0 = 3$$

Optimal alignment score = 3

Smith-Waterman Algorithm (1)

- ❖ When the score drops to zero, extension of path is terminated and a new one can begin
- ❖ There can be many individual paths bounded by regions of poorly matching sequence; the one with the highest score is reported as the optimal local alignment
- ❖ Optimal methods always report the best alignment that can be achieved, even if it has no biological meaning

Smith-Waterman Algorithm

- ❖ A simple modification to the basic strategy allows the optimal local alignment to be found
- ❖ The path for this alignment does not need to reach the edges of the search graph but may begin and end internally
- ❖ An alignment would be locally optimal if its score cannot be improved either by increasing or decreasing the extent of the alignment
- ❖ The cumulative score for a path will decrease in regions of poorly matching sequences

Smith-Waterman Algorithm (2)

- ❖ Semiglobal alignment is not sufficient for searching best match subsequences
- ❖ Because mismatching positions and gaps outside target subsequences produce favourable score
- ❖ In addition to the 3 basic (vertical, horizontal and diagonal) movements, there is a forth option
 - ✧ Score zero if all other movements have negative

Smith-Waterman Algorithm (3)

- ❖ Initialised the first row and column of the table with zeros
- ❖ Fill in all partial scores
- ❖ Find the maximum partial score and work backward until a zero score is reached

Initialisation Step

- ❖ Create Matrix with $M + 2$ columns and $N + 2$ rows.
- ❖ The first row and column of the table are signed with zeros

		A	G	A	T	C	A	T
	0	0	0	0	0	0	0	0
A	0							
G	0							
A	0							

Example 6

Find the optimal alignment for the sequences AGATCAT and AGA, match +1, mismatch 0 and gap -1

Calculation of Scores

- ❖ Match = +1 , Mismatch = 0, Gap = -1 or zero

		A	G	A	T	C	A	T
	0	0	0	0	0	0	0	0
A	0	1	0	1	0	0	1	0
G	0	0	2	1	1	0	0	1
A	0	1	1	3	2	1	1	0

Traceback Path

- ❖ Find the maximum partial score and work backward until a zero score is reached

		A	G	A	T	C	A	T
	0	0	0	0	0	0	0	0
A	0	1	0	1	0	0	1	0
G	0	0	2	1	1	0	0	1
A	0	1	1	3	2	1	1	0

Optimal Local Alignment

A G A T C A T
 | | |
A G A

$$1 + 1 + 1 + 0 + 0 + 0 + 0 = 3$$

Optimal alignment score = 3